INTRODUCTION

The information in this instruction will help guide Field Engineers through the assembly and installation of the three-piece Lucas Airo Space (Bendix) Load Coupling. The method described in this article does not supersede any factory instructions.

REFERENCE DOCUMENTS

No.	Description
ML0411	Field Alignment Instructions
EM5315	MS6001B Installation Procedure
DLN-048	Generator Assembly Diagram
DLN-064	Generator Mechanical Outline
DLN-060 and 065	Generator Package Wiring

SPECIAL TOOLS

Special turning tools required for assembly (field fabricated) are shown in Figures 1 and 2.

THREE-PIECE LOAD COUPLING INSTALLATION

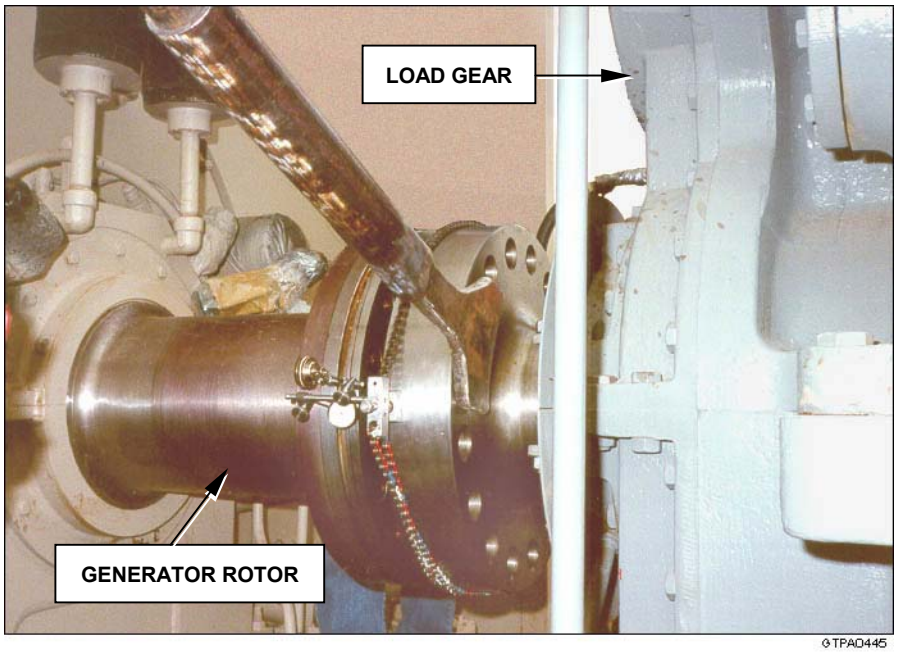
Remove the Bentley Vibration Probes and conduit from the aft side (generator end) of the Load Gear.

The Generator Field should already be floated back toward the collector end (0.750 inch per ML 0411 instruction), so that a measurement of 31-1/2 inches is obtained from the generator front bearing centerpunch marks (on bearing cap or on side of bearing housing) to the face of the generator field. See Figure 3 and Figure 4.

Shrink the rabbit fit by packing the turbine end coupling adapter plate in dry ice overnight. The turbine rotor hub and adapter plate has a 0.004 to 0.005 inch interference fit. Assemble the plate to the turbine rotor. Torque all bolts to specification (see MLI 0411).

Turn the load gear over and float both gear shafts aft (toward the generator) until a measurement of 74-1/2 inches is obtained between the adapter plate bolted onto the turbine rotor and the face of the load gear. See Figure 1 and Figure 5.

Note: Once it has been started the load gear will turn by hand, and will only float downstream 9/16 inch from the center. (See Figure 3.)



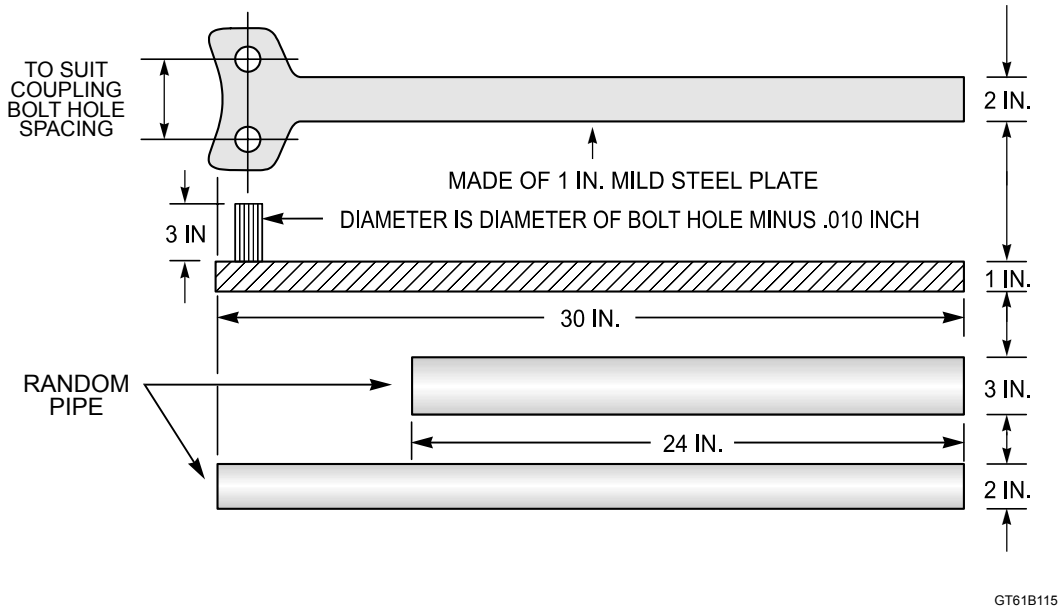


Figure 1. Load Gear Coupling Hub Wrench – Typical Layout (used to turn over either the Generator Rotor or Load Gear for alignment readings)

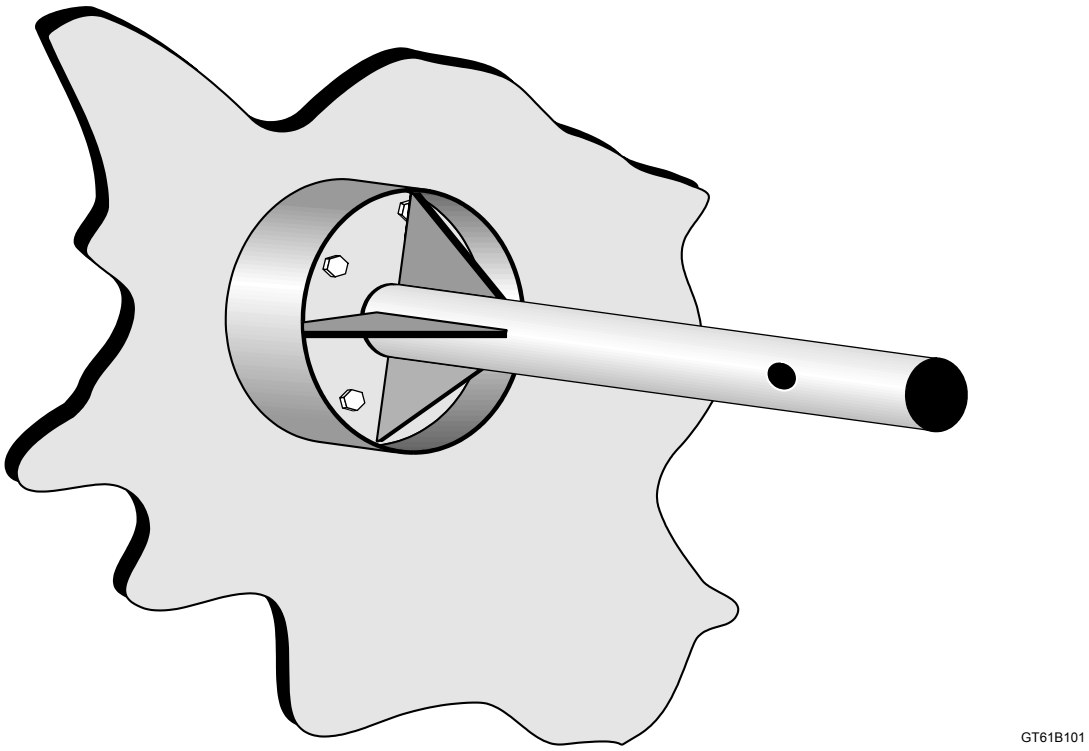


Figure 2. Turbine Rotor Turning Fixture – Typical Layout

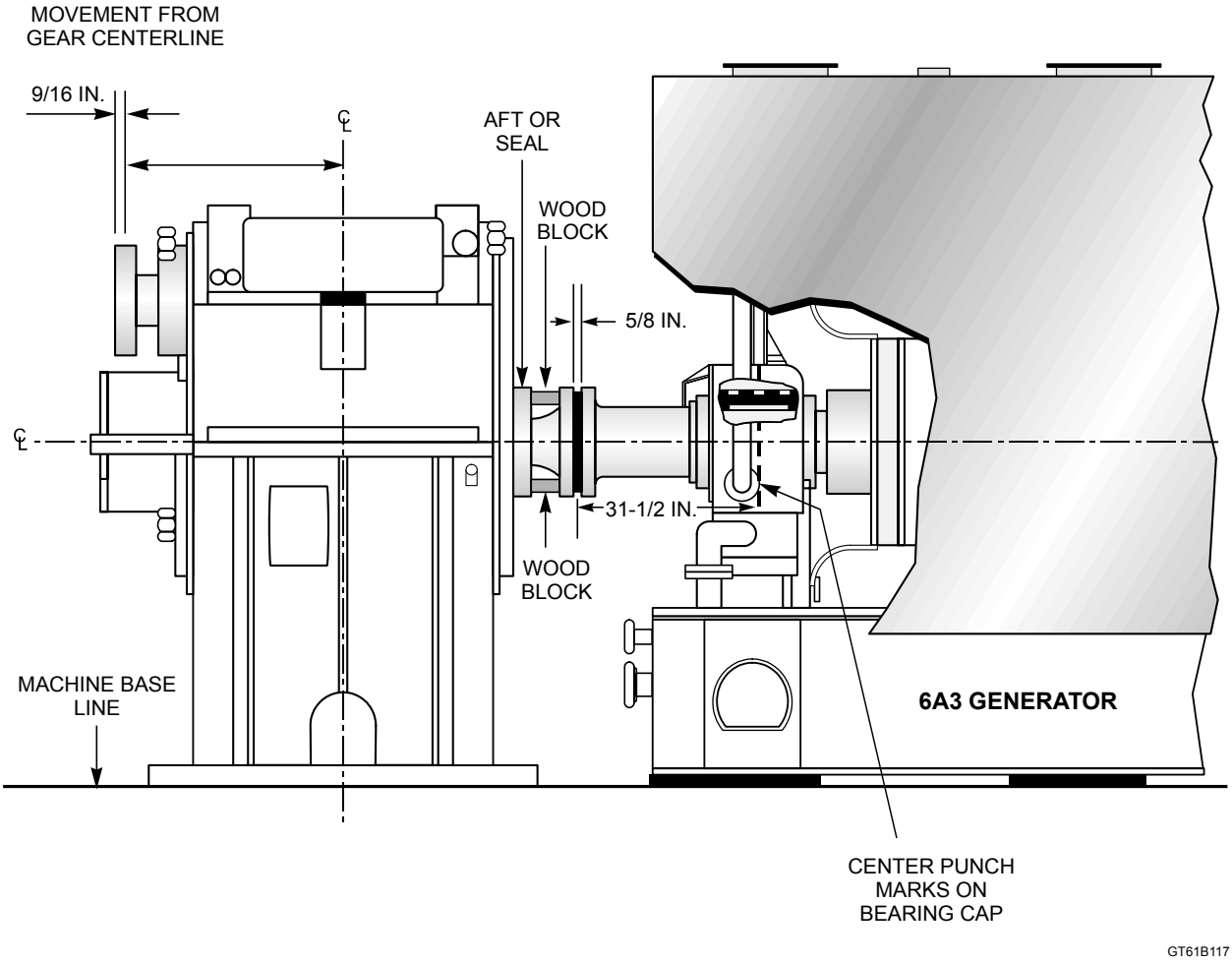


Figure 3. 6A3 Generator, Load Gear Matchup – Typical Layout

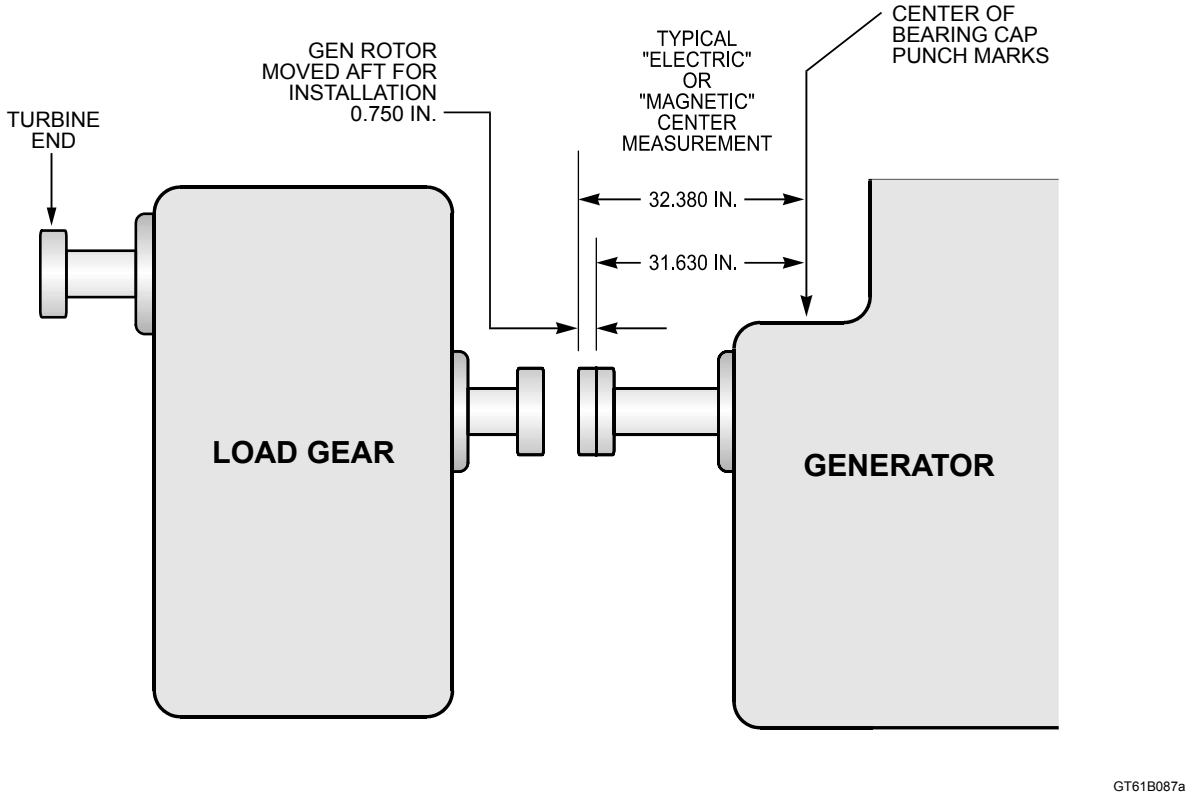


Figure 4. Generator Installation Field Position Data

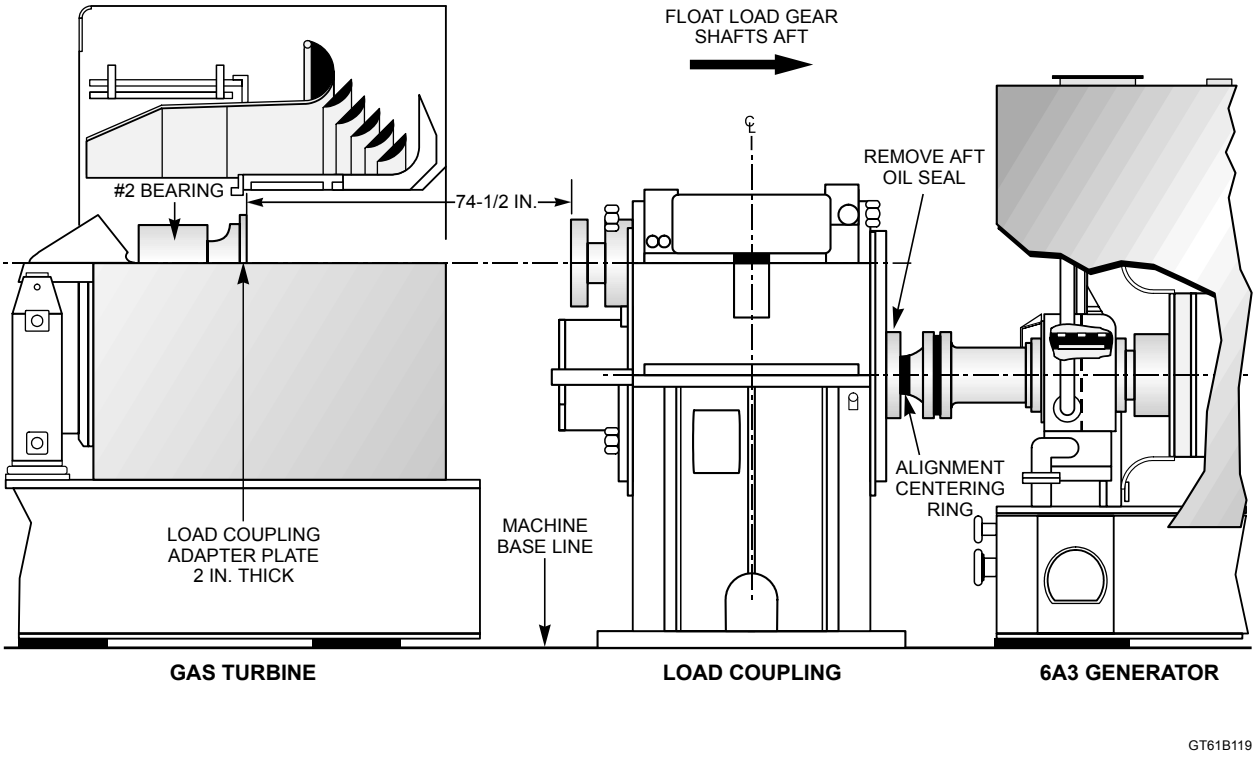


Figure 5. Load Coupling Installation – Typical Layout

Rig and set coupling center (spool) piece down into the turbine load tunnel (see Figure 6).

- If the load gear enclosure is not set over the gear: rig one end of a long nylon strap to the coupling (spool) piece and the other end to a picker. Set the coupling down into the load tunnel.

- If the load gear enclosure is set over the gear: rig a nylon strap around the two air ejector pipes running up the aft side of the exhaust plenum. Assemble a 1/2-ton come-a-long to the strap, and another long nylon strap to the coupling centerpiece and come-a-long. Set the coupling down into the load tunnel.

Shrink the rabbit fit by packing the load gear end coupling adapter plate in dry ice overnight. The gear hub and adapter plate has a 0.004 inch to 0.005 inch interference fit. Assemble the plate to the load gear hub.

Move the spool piece side to side so that the coupling bolts can be assembled into the gear hub through the adapter plate. Torque all bolts to specification (refer to MLI 0411).

Set the load coupling spoolpiece in position on its match marks. Assemble the turbine end bolts snuggly (do not torque at this time).

Float the load gear shafts back upstream so that the match marks on the coupling adapter (bolted to the load gear) match up with the marks on the spoolpiece. Assemble coupling bolts snuggly.

Note: A turning fixture (field fabricated) is bolted to the front of the turbine rotor so that the rotor can be turned over to assemble and torque the load coupling bolts. (See Figure 2.)

Float the generator field upstream by turning the field while jacking with a port-a-power. The jack should be placed between two pieces of hardwood behind the generator coupling hub (see Figure 7).

- Another method is to assemble two (2) pieces of all thread through the bolt holes in both hubs (180 degrees apart). Assemble nuts to the all thread and draw the field upstream.

At this point refer to MLI 0411, Basic Field Alignment Instructions, for assembly of the generator to load gear coupling bolting.

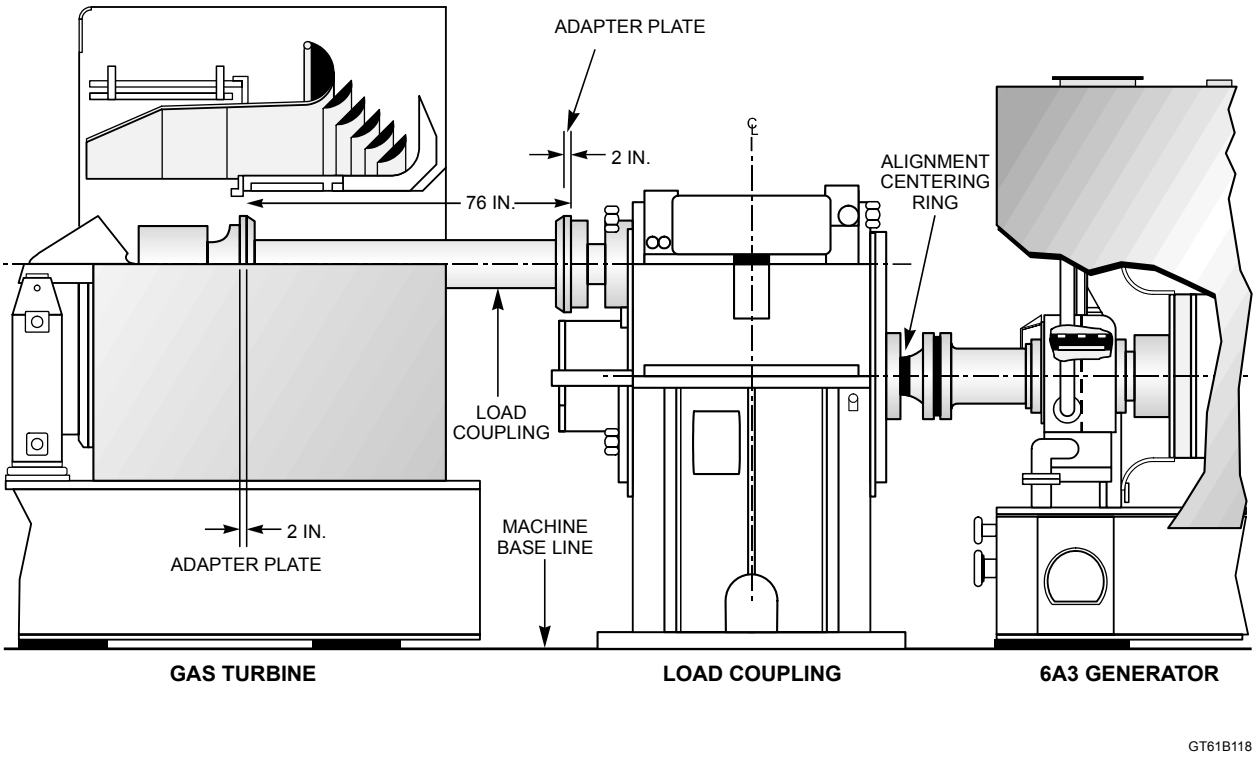


Figure 6. Load Coupling Installation – Typical Layout

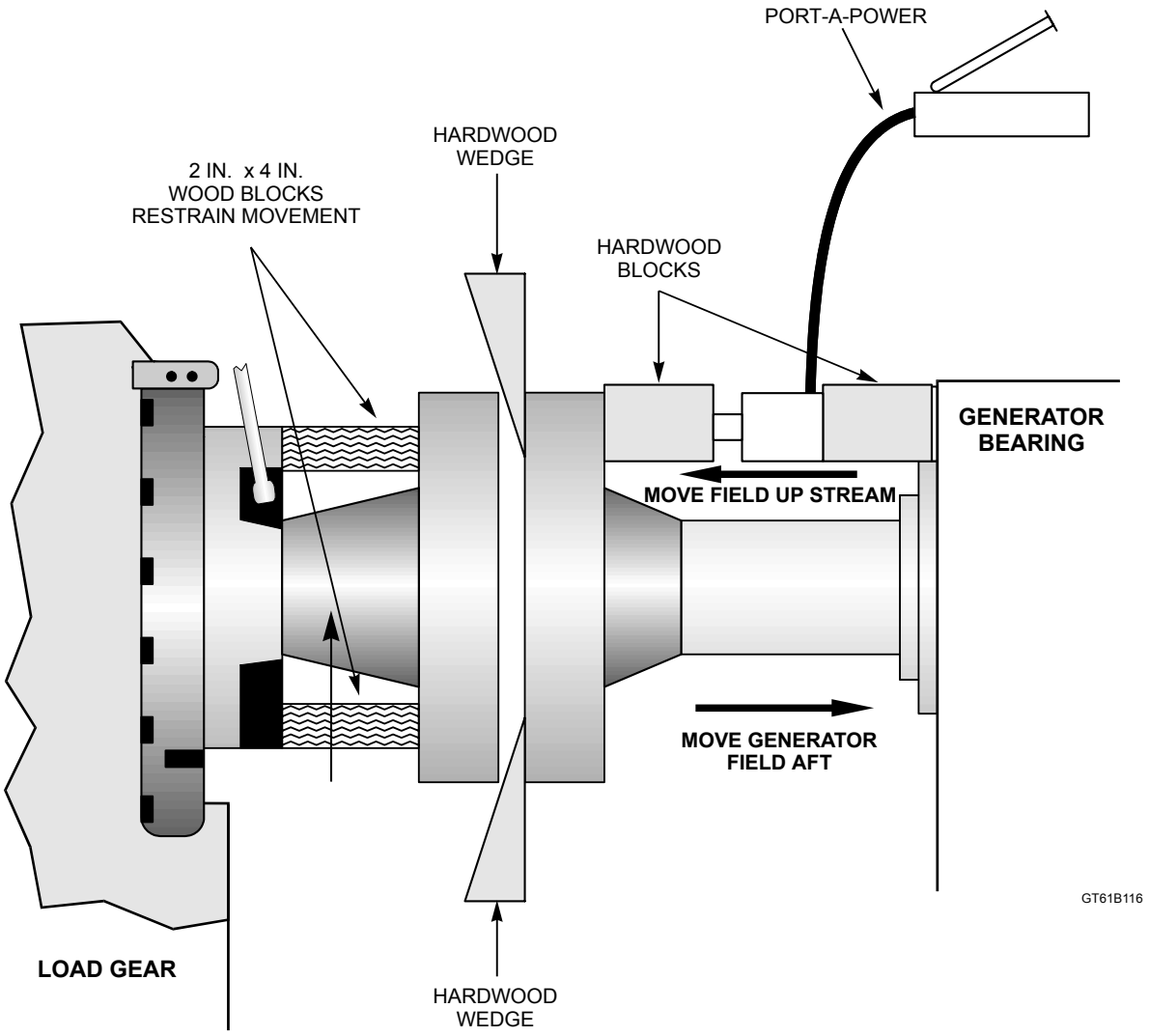


Figure 7. Rigging to Float Generator Field – Typical Layout